

e-Voting Machine

Investor Presentation

S4Y Solutions

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The challenges of voting online

- **Verifiable and secret at once.** A voter wants proof that their vote counted; a secret ballot means nobody may learn how they voted.
- **No voting booth.** At home, a voter can be watched, pressured or paid to vote a certain way, and asked to show proof.
- **Identity.** Each eligible voter must get exactly one say, with a predictable level of certainty about who they are.

Published work has tackled this for twenty years

Estonia's i-voting, Helios, JCJ/Civitas and MACI each solve part of it, at the cost of complex cryptography that is hard to explain to the people who vote.

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The idea: revocable votes on a public, add-only ledger

Change your vote at any time while the voting is open

A voter can cancel their vote and vote again. Every vote is a public `+1`; a cancellation is a public `−1` that is **not linked** to the vote it cancels.

For the voter

A receipt finds their vote on the public ledger, so they can check it was counted as cast.

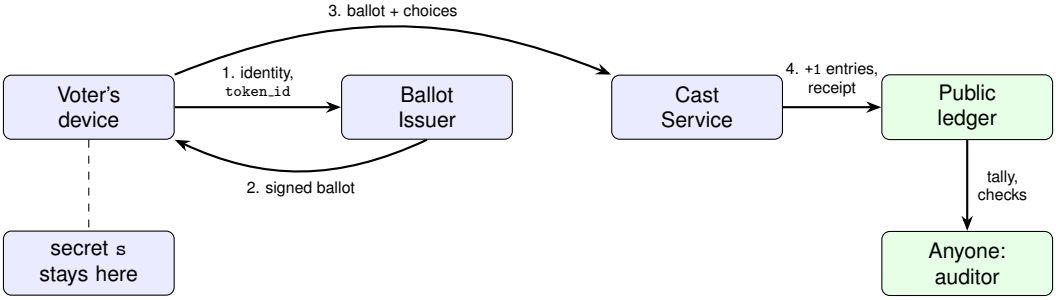
Against a coercer

The receipt proves the vote was cast, not that it still counts. The voter can change it later, alone, and the coercer cannot tell.

The tally is the sum of the ledger: anyone can add it up.

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How it works



- `token_id = H("token" | s)`, receipt `= H("cast" | s)`: both derived from a secret only the voter holds.
- **Re-vote:** the voter proves the old vote with `s`; the ledger gets `-1` entries with no receipt, and the voter a new ballot.
- Every entry is signed and chained to the previous one by its hash: nothing can be rewritten unnoticed.

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The building blocks

Ballot token	Signed by the Issuer. One first ballot per eligible voter; a re-vote ballot only in exchange for a cancelled vote
Receipt	The cast ID. Finds the voter's `+1` entries on the ledger
Add-only ledger	`+1` and `−1` entries, signed by the operator and hash-chained. Nothing is ever edited or deleted
Voter registry	Public: random pseudonyms and whether each one received a ballot. Never who they are
"Smart contract"	The published, deterministic rule that checks and applies each entry. No blockchain is required
Readable signatures	Every signed value is shown as plain text plus its real signature tag, so a tampered line visibly fails

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What anyone can check

	Voter	Anyone	Operator
My vote was recorded as I chose	✓		
The tally equals the sum of the ledger	✓	✓	
The ledger was not rewritten (signatures, hash chain)	✓	✓	
No more net votes than ballots issued	✓	✓	
Nobody took a ballot in my name	✓		
Every cancellation was made by the vote's owner			trusted
The issuer does not link my identity to my ballot			trusted

The last two rows are the honest limits of the first version. The roadmap removes them.

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Try to cheat: live in the demo

Each attempt runs against the real system and is rejected with its reason:

- Ask for a second ballot — rejected: already issued
- Cast the same ballot twice — rejected: token already spent
- Forge the issuer's signature on a ballot — rejected: not signed by the issuer
- Cancel someone else's vote without its secret — rejected: no matching vote
- Cancel the same vote twice — rejected: already cancelled
- Edit a ledger line — its signature turns red

After every attempt, the public auditor re-runs its checks and they all still pass.

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Where this stands among existing systems

System	Against coercion	Difference
Estonia i-voting	Re-voting; the last vote counts	Encrypted votes; earlier ones are dropped at tally time
Helios, Belenios	None: a voter can prove their vote	Encryption and trustees; hard to explain
JCJ, Civitas	Fake credentials	Strongest model; heavy tally, hard to use
MACI	Re-voting and key changes	Encrypted messages, proofs of a correct tally
This project	Re-voting inside the ledger	Public ±1 entries: a tally anyone can add up

Stand-out: explainability, re-voting without a separate tally step, an honest staged trust model, and pluggable identity.

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Business model

- **Voting as a service:** a secure, transparent cast ledger for associations, unions, shareholder meetings and universities.
- **Identity plugins:** paid identification, from email with risk scoring to bank IDs, eID and organisation directories.
- **Related services:** live turnout, historical data API, private deployments.

The first market is still open (associations and unions, shareholders, universities, DAOs); it decides the identity plugins and the trust story to lead with.

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Roadmap: removing the remaining trust

	Step	Removes
RM1	Blind-signature ballots and an anonymous channel	The issuer linking a voter to a ballot
RM2	Zero-knowledge cancellations with public nullifiers	The operator faking or repeating cancellations
RM3	Ledger head anchored on a public chain or log	Showing different histories to different people
RM4	Threshold keys, several independent operators	Any single operator acting alone

Also after investment: real identity providers, persistence and scaling, an external security audit.

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Current status

- **Working demo** in the browser, with no backend: organiser, voter, public ledger and auditor, and the “try to cheat” panel. Published with GitHub Pages.
- **Core domain** in Scala 3: voting, voter registry, ballot issuance, casting, re-voting, the signed hash-chained ledger and the auditor, covered by automated tests.
- **Design documents**: requirements, threat model, data model, flows, trust model, prior art.

`https://github.com/s4ysolutions/e-voting-machine`

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